

1. Let cheapY = (Fun f →  
 Let bot = Fun X → ((Fun y → y) (Fun y → y)) In  
 Let f10 = f (Fun X → bot X) In  
 Let f9 = f (Fun X → f10 X) In  
 Let f8 = f (Fun X → f9 X) In  
 Let f7 = f (Fun X → f8 X) In  
 Let f6 = f (Fun X → f7 X) In  
 Let f5 = f (Fun X → f6 X) In  
 Let f4 = f (Fun X → f5 X) In  
 Let f3 = f (Fun X → f4 X) In  
 Let f2 = f (Fun X → f3 X) In  
 Let f1 = f (Fun X → f2 X) In  
 f1)";

2. a. (Fun x → x+1)  
 $\cong$  (Fun y → (x+1 [y/x])) ( $\alpha$ -equivalence)  
 $\cong$  (Fun y → (y+1)) (substitution)  
 $\cong$  (Fun a → (Fun y → y+1) a) ( $\eta$ -equivalence)  
 So (Fun x → x+1)  $\cong$  (Fun a → (Fun y → y+1) a) By (Transitivity)

b. (Let x = f 1+2 In 0)  
 $\cong$  0 [(f 1+2)/x] (Let- $\beta$ )  
 $\cong$  0 (substitution)  
 $\cong$  1-1 (the difference equivalence for concrete integer values)  
 So (Let x = f 1+2 In 0)  $\cong$  1-1 By (Transitivity)

c.  $C \stackrel{\text{def}}{=} (\text{Let } x = \text{False in Let } y = \text{True in Let } f = (\text{Fun } z \rightarrow 0) \text{ in } (\bullet) 1)$   
 we have the left hand side:

Let x = False in Let y = True in Let f = (Fun z → 0) in  
 (Fun x → If y Then Fun y → x+y Else f x) 1),

which evaluates to Fun y → 1+y.

for the right hand side, we have:

Let x = False in Let y = True in Let f = (Fun z → 0) in

(Fun y → If x Then Fun x → y+x Else f y) 1),

which evaluates to 0.

The two results are of different types, so it's not an observational equivalence.

d. (Fun a → Fun b → a+b)  
 $\cong$  (Fun c → (Fun b → a+b [c/a])) ( $\alpha$ -equivalence)  
 $\cong$  (Fun c → (Fun b → c+b)) (substitution)  
 $\cong$  (Fun c → (Fun a → (c+b [a/b]))) ( $\alpha$ -equivalence)  
 $\cong$  (Fun c → (Fun a → (c+a))) (substitution)  
 $\cong$  (Fun b → ((Fun a → (c+a)) [b/c])) ( $\alpha$ -equivalence)  
 $\cong$  (Fun b → (Fun a → (b+a))) (substitution)  
 So (Fun a → Fun b → a+b)  $\cong$  (Fun b → Fun a → b+a) By (Transitivity)

3. a. Variant- $\beta$  :

Match  $n_j(v_j)$  with

$$|n_1(x_1) \rightarrow e_1| \dots \cong e_j[v_j/x_j]$$

$$|n_j(x_j) \rightarrow e_j| \dots$$

$$|n_m(x_m) \rightarrow e_m| \dots$$

b. Match 'Red (Fun  $x \rightarrow x$ ) with

$$| \text{'Red } f \rightarrow \text{'Green } (f \ 5)$$

$$| \text{'Green } m \rightarrow \text{'Red } (\text{Fun } x \rightarrow x + y)$$

$$\cong \text{'Green } (f \ 5) [(\text{Fun } x \rightarrow x) / f] \quad (\text{Variant-}\beta)$$

$$\cong \text{'Green } ((\text{Fun } x \rightarrow x) \ 5) \quad (\text{substitution})$$

$$\cong \text{'Green } (x \ [5/x]) \quad (\beta\text{-Equivalence})$$

$$\cong \text{'Green } (5) \quad (\text{substitution})$$

4. a. type mypair = { fst: 'a; snd: 'b } ::

let Pair (x: 'a) (y: 'b): ('a, 'b) mypair = { fst = x; snd = y } ::

let Fst (p: ('a, 'b) mypair): 'a = p.fst ::

let Snd (p: ('a, 'b) mypair): 'b = p.snd ::

b. Let  $p = \text{pair}(\text{Int } 1, \text{pair}(\text{Int } 2, \text{Int } 3))$  In

Equal (Fst (Snd (p)), Int 2)

5. a.

$$\frac{\overline{\text{'No}(3) \Rightarrow \text{'No}(3)} \quad \overline{y \ [3/y] \Rightarrow 3}}{\text{Match 'No}(3) \text{ with 'Yes}(x) \rightarrow 3 \mid \text{'No}(y) \rightarrow y \Rightarrow 3}$$

$$\frac{(\text{Fun } y n \rightarrow \dots) \Rightarrow \star \quad \overline{\text{'No}(3) \Rightarrow (\text{'No}(3))} \quad \text{Match } y n \text{ with 'Yes}(x) \rightarrow 3 \mid \text{'No}(y) \rightarrow y \ [ \text{'No}(3) / y n ]}{(\text{Fun } y n \rightarrow \text{Match } y n \text{ with 'Yes}(x) \rightarrow 3 \mid \text{'No}(y) \rightarrow y) \ (\text{'No}(3)) \Rightarrow 3}$$

$$\text{Fun } r_2 \rightarrow \text{If } r_1.c \text{ Then } r_1.a \text{ Else } \{a = r_2.c\}$$

b.

$$\frac{\overline{\{a=1; C=\text{True}\} \Rightarrow \star} \quad \overline{\{a=1; C=\text{True}\} \Rightarrow \star}}{\overline{\{a=1; C=\text{True}\}.C \Rightarrow \text{True} \quad \{a=1; C=\text{True}\}.a \Rightarrow 1}}{\text{If } \{a=1; C=\text{True}\}.C \text{ Then } \{a=1; C=\text{True}\}.a \text{ Else } \{a=\{a=\text{False}; C=2\}.c\} \Rightarrow 1}$$

$$\frac{(\text{Fun } r_2 \rightarrow \dots) \Rightarrow \star \quad \overline{\{a=\text{False}; C=2\} \Rightarrow \star} \quad \text{If } \{a=1; C=\text{True}\}.C \text{ Then } \{a=1; C=\text{True}\}.a \text{ Else } \{a=r_2.c\} \ [ \{a=\text{False}; C=2\} / r_2 ]}{\text{Fun } r_2 \rightarrow \text{If } \{a=1; C=\text{True}\}.C \text{ Then } \{a=1; C=\text{True}\}.a \text{ Else } \{a=r_2.c\} \Rightarrow 1}$$

$$\frac{(\text{Fun } r_1 \rightarrow \dots) \Rightarrow \star \quad \overline{\{a=1; C=\text{True}\} \Rightarrow \star} \quad \text{Fun } r_2 \rightarrow \text{If } r_1.c \text{ Then } r_1.a \text{ Else } \{a=r_2.c\} \ [ \{a=1; C=\text{True}\} / r_1 ]}{(\text{Fun } r_1 \rightarrow \text{Fun } r_2 \rightarrow \text{If } r_1.c \text{ Then } r_1.a \text{ Else } \{a=r_2.c\}) \ \{a=1; C=\text{True}\} \ \{a=\text{False}; C=2\} \Rightarrow 1}$$